

SC444 Game Theory (Winter 2026)

Instructor: G Venkatesh, Director's office

Course Placement: B.Tech. (ICT, ICT-CS, MnC), Semester-VI

Course format: 3 Lecture hours per week

What is game theory?

Game theory analyzes situations in which the decisions and actions of two or more individuals (or firms, political parties, countries) are interdependent. The foundational assumption is that decision-makers are **rational** and reason **strategically**—meaning their reasoning takes into account their knowledge and expectations of other decision-makers. While originally developed for economics, Game Theory has proven essential for understanding conflict and cooperation in management, political science, law, biology, and recently, in the design of agentic AI systems.

Methodology:

Delivered through conceptual lectures, supported by Jupyter notebook model demos and in-class game sessions.

- **Course Project:** This creates hands-on experience in identifying game-like problem situations, creating an evolutionary game model for them, and using the model to devise remedial measures.
- **Logistics:** Students must enrol in **Google Classroom** to access course materials and instructions. All communication will occur via this platform.
- **Technical Workflow:** Project submissions must be Jupyter notebooks coded in **Python** and executed using **Google Colab** within the Google Classroom/Drive environment.

Prerequisites:

- **Essential:** Comfort with logical reasoning, basic probability, and coding in Python.
- **Desirable:** Some background in intermediate microeconomics.

Evaluation plan:

Grading will be calculated based on the following components:

- **Group Project:** 30%
- **Quizzes:** 30% (Based on the best 3 out of 4 quizzes)
- **End-term Exam:** 40%

Course Outcomes:

This course aims to provide a systematic introduction to the tools of game theory and its applications. By the end of the course, students will be able to create mathematical and computational models to understand:

- **Market Dynamics:** Competition among firms, Duopolies and oligopolies
- **Strategic Behaviour:** Credible deterrence strategies, Benefits of being unpredictable, Negotiation and bargaining, Strategic voting.
- **Information Economics:** Eliciting information, Principal-agent games, Moral hazard in insurance, Signaling games and advertising, Adverse selection and the use of warranties.
- **Mechanism Design:** Incentive design, Auction design, Product bundling.
- **Evolutionary Models:** The evolution of cooperation and free-rider elimination.

Students will demonstrate these outcomes by developing their own evolutionary game models for specific situations, documented as Jupyter notebooks.

Text Books:

"A primer in game theory" by Robert Gibbons, Pearson India, 2018.

"A Course In Game Theory" by Martin Osborne and Ariel Rubinstein, MIT Press, 1994

<https://sites.math.rutgers.edu/~zeilberg/EM20/OsborneRubinsteinMasterpiece.pdf>

Reference Books:

"An introduction to Game Theory" by Martin Osborne, Oxford University Press, 2012

"Game Theory" by Giacomo Bonanno, https://faculty.econ.ucdavis.edu/faculty/bonanno/PDF/GT_book.pdf

"Fun and games: A text on Game theory" by Ken Binmore, (Indian ed) AITBS Publishers, New Delhi, 1999.

"Game theory with economic applications" by H S Bierman and L Fernandez, (Indian ed), Pearson education, 1998.

"Games and information" by Eric Rasmusen, Basil Blackwell Ltd. 1989.

"Strategies and Games: Theory and Practice" by Prajit K Dutta, MIT Press.

Detailed Course Contents		
Topic Name	Content	No. of weeks
Basic strategic form games and solutions	• Co-operation, Co-ordination, Confrontation, Location, Pricing games • Zero sum and non-zero sum games • Iterated Eliminated of Dominated Strategies • Nash equilibrium • Duopoly and Oligopoly • Mixed strategies and the benefits of being unpredictable	4
Multi stage games	• Repeated games and the evolution of co-operation • Evolutionary games • Commons problem and remedies • Strategic Voting • Extensive form games and backward induction • Credible deterrence • Chain store paradox and the Centipede game • Bargaining	5
Incomplete information	• Asymmetric duopoly • Principal-Agent and incentive design • Adverse selection • Signaling games • Mechanism design • Product bundling • Auction design	5
Group Projects	Students will form into groups of 3-5, pick a problem, model it as a game and submit it as a Jupyter notebook	